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Separation distress in infant rhesus monkeys: effects of diazepam and Ro 15-1788

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Disruption of the primate mother–infant attachment bond is a naturally occurring stressor that results in marked behavioral, physiological, and endocrine activation. We studied the effect that altering benzodiazepine systems has on the behavioral and endocrine response of infant rhesus monkeys (1–27 weeks of age) to brief separation from their mothers. In the first experiment, the benzodiazepine agonist diazepam (0.1 and 1.0 mg/kg) significantly increased locomotion and social behavior and decreased inactivity and distress vocalizations in infant monkeys undergoing separation. In the second experiment, the benzodiazepine antagonist Ro 15-1788 (5 and 10 mg/kg) had no significant effects on the infants' separation response. In the third experiment, administration of diazepam 1.0 mg/kg was followed by administration of Ro 15-1788 10 mg/kg in infants undergoing separation. Ro 15-1788 blocked the decreases both in inactivity and in plasma ACTH and cortisol concentrations caused by diazepam, suggesting that these effects are mediated through benzodiazepine receptors. These data support the hypothesis that in primates, endogenous benzodiazepine systems modulate the behavioral and endocrine response to the naturally occurring stress of separation.

INTRODUCTION

The disruption of attachment bonds is a stressful event that occurs frequently in the lives of human and non-human primates[3,20,25,27,29,40] and other social species. Its consequences depend on a variety of neurobiological and environmental factors. Infants are generally most vulnerable, as their very survival depends on the presence of their care-givers[17,20,35,37,40].

An infant primate's response to separation from its mother is immediate and dramatic. It becomes behaviorally agitated and emits species-specific distress vocalizations. This intense behavioral response is important for survival, since it maximizes the mother's chances of relocating her lost infant. Laboratory studies in non-human primates show that separation induces autonomic[34] and endocrine[5,19,26] activation concomitant with the behavioral response. These physiologic changes provide metabolic support for the infant while its mother is absent.

The response of non-human primate infants to separation is very similar to that of human infants. In now classic studies on young children placed into non-supportive environments, Bowlby[3] and Spitz[39] described serious effects, both acute and long-term, of separation.

Recent studies suggest that the endogenous benzodiazepine system plays an important role in regulating behavioral and physiological arousal[1,6,7,9,23]. Benzodiazepine receptors occur throughout the central nervous system[21,43] and putative endogenous benzodiazepine receptor ligands have been identified[1,36]. Benzodiazepine receptor ligands are powerful neurophamacologic tools for studying anxiety-related behaviors in various animal models[2,4,6,8,9,11,12,32,33,38,42] and their effects on anxiety have also been established in humans[10,22]. In this paper we report studies designed to establish whether specific neuropharmacological manipulations of brain benzodiazepine systems modulate the infant rhesus monkey's re-

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sponse to brief separation. If they do, this would support the hypothesis that endogenous benzodiazepine systems are involved in mediation of the primate's response to separation.

In the first experiment we administered the benzodiazepine agonist diazepam, which has clear anxiolytic effects in humans[22], to monkeys undergoing separation. In animals, benzodiazepine agonists reduce the suppressive effects of punishment on responding, increase exploration in novel environments, and increase social interaction[4,8,11,41,42]. We anticipated that diazepam would decrease separation-induced distress in infant monkeys.

In the second experiment we administered Ro 15-1788 to monkeys undergoing separation. Ro 15-1788, an imidazodiazepine, binds to the benzodiazepine receptor and prevents the effects of other compounds that work through benzodiazepine receptors. Initially, Ro 15-1788 was thought to be a true benzodiazepine antagonist, but recent work has demonstrated that it has its own intrinsic actions[16]. In some paradigms, Ro 15-1788 has anxiogenic effects when administered in low doses[13,15] and anxiolytic effects at higher doses[16]. The purpose of this experiment was to identify any intrinsic effects of Ro 15-1788 on the infant monkeys' response to brief separation and to establish a dose of Ro 15-1788 that has no intrinsic effects, making it useful in later experiments attempting to block the effects of diazepam.

In the third experiment, diazepam was administered with Ro 15-1788. If Ro 15-1788 blocked the effects of diazepam, this would establish that the effects of diazepam are mediated through benzodiazepine receptors. Since we and others have demonstrated pituitary–adrenal activation induced by separation in rhesus monkeys[19,26], we sampled blood for determination of ACTH and cortisol. Based on other work[14,18,30], we hypothesized that the concentrations of these stress-related pituitary–adrenal hormones would be decreased by diazepam.

MATERIALS AND METHODS

Subjects

The experimental subjects were 8 infant rhesus monkeys (*Macaca mulatta*), 6 males and 2 females, ranging in age from 1 to 27 weeks and in weight from 0.55 to 1.25 kg. They were housed with their mothers in cages 71.5 cm wide by 71.5 cm high by 68 cm deep, and were breast-fed by their mothers. The monkeys were maintained on a 12-h light–dark cycle (lights on from 06.00 to 18.00 h).

Experimental design

A within-subjects design, counterbalanced for the order of drug administration, was used. Each animal received each drug dose.

Drug dosage and testing procedures

In the first experiment, each animal received vehicle and diazepam, 0.1 and 1.0 mg/kg. These doses were selected on the basis of data from experiments in older rhesus monkeys (unpublished data). The vehicle consisted of 40% propylene glycol, 10% ethanol, 1.5% benzyl alcohol, and 48.5% water (v/v), with benzoic acid 1.2 mg/ml and sodium benzoate 48 mg/ml (Hoffmann-La Roche). The drug was administered in a concentration of 5 mg/ml. The 3 treatments were separated by intervals of 1 week.

On each test day, 4 infants were given diazepam or vehicle intramuscularly at 09.20 h while they were with their mothers. Forty min later they were separated from their mothers and placed in another room, in individual sections of a larger cage that had been subdivided into 4 cages (each measuring 75 cm high by 75 cm wide by 60 cm deep) by quadrasecting mesh panels. This arrangement allowed the infants to contact each other with their fingers if they desired. At 10.20 h, behavioral testing was initiated by a rater who was unaware of the treatment conditions. Each infant was tested for two 5-min intervals, with 30 min between test sessions. One hour after separation, the infants were reunited with their mothers.

In the second experiment, two doses of Ro 15-1788 (gift from Hoffmann-La Roche, 5 and 10 mg/kg) dissolved in a vehicle solution were tested, with vehicle serving as a control. The vehicle consisted of Emulphor (GAF Corp., NY), ethanol, and 0.9% NaCl, 1:1:2 by volume. For the 5 mg/kg dose, the drug was administered in a concentration of 5 mg/ml; for the 10 mg/kg dose, it was 10 mg/ml.

At 10.00 h each mother–infant pair was handcaught and separated. The drug was administered intramuscularly within 10 min of separation, and behavioral testing was initiated 5 min after drug administration. Because of the short half-life and duration

of action of the drug, drug administration and behavioral testing could be repeated with as little as 48 h between treatments.

In the third experiment, each treatment consisted of two injections. The first, given at 09.20 h, consisted of either diazepam 1 mg/kg or the diazepam vehicle solution. The second, given at 10.10 h, was either Ro 15-1788 10 mg/kg or the vehicle used to dissolve it. We administered the drugs in this sequence in order to synchronize the peak effects of the two drugs. Immediately after the second injection, the infants were separated from their mothers. Behavioral testing began at 10.15 h. At 11.15 h the infants were caught and anesthetized with ketamine HCl 10 mg/kg, and 2 ml of blood was collected by femoral venipuncture[24]. Animals were tested at weekly intervals, each one eventually receiving each combination of drugs: vehicle/vehicle, diazepam/vehicle, vehicle Ro 15-1788, and diazepam/Ro 15-1788.

Behavioral scoring and analysis

The behavioral scoring and software system was developed at the Harlow Primate Laboratory[28]. It utilizes a syntactic encoding process that preserves the frequency, duration, coincidence, and sequence of behaviors. The behavioral data collection apparatus is an APPLE II+ microcomputer. Our present scoring system uses 35 operationally defined behavioral scoring categories. The behaviors most frequently scored in this study are defined in Table I.

Data for each behavior were analyzed by a repeated measures ANOVA, in which factor A was dose of drug and factor B was test session. Duncan's *t*-tests were used for specific post hoc comparisons.

Sample preparation and radioimmunoassays

Within 8 min of ketamine administration, all samples were collected into tubes containing EDTA and immediately placed on ice. Plasma was separated from blood by centrifugation at 3000 rpm at 4 °C for 10 min. Plasma was then frozen at −70 °C until assayed.

ACTH immunoreactivity was measured in unextracted plasma with a completely characterized antiserum (IgG-ACTH-1, IgG Corporation, Nashville, TN)[24,31]. The sensitivity of the assay is 0.3 pg/tube. The intra-assay and interassay coefficients of variability are 4.2% and 14.6%, respectively. Serial dilutions of monkey plasma result in parallel displacement of the radioactive tracer compared with the standard $ACTH_{1-39}$ (ref. 24).

Plasma cortisol was measured with reagents obtained from New England Nuclear (Boston, MA). The sensitivity of the assay is 2 μg/dl, with an intra-assay variability of 2.9% and an interassay variability of 4.8%.

TABLE I

Behavioral categories showing alterations due to drug treatments in infant rhesus monkeys

Category	*Description*
Environmental exploration	Any manual or oral exploration or manipulation of the physical surroundings (cage, pan, droppings, playing with chow, etc.). Includes non-locomotive cage shake.
Locomotion	Ambulation of one or more full steps at any speed. Includes such behavior as dropping from ceiling to floor or swinging cage shake. May be scored with other behaviors.
Vocalizations	Any audible communication.
Inactivity	Maintenance of stable, stationary position with no simultaneous scorable behavior except proximity and vocalization.
Self-mouth	Oral contact with any part of own body, excluding self-aggression, self-groom, and self-sex.
Social behavior	Maintenance of a distance of less than 15 cm to another animal; may or may not include manual contact or manipulation through the wire mesh.

RESULTS

In the first two experiments, animals that received vehicle displayed an intense separation response characterized by frequent vocalizations and agitated behavior. In Expt. 1, the general effects of diazepam were to increase activity and social behavior while decreasing distress vocalizations (Fig. 1, A–C). Locomotion increased from a mean ± S.E.M. of 197.2 ± 19.0 s after vehicle to 216.5 ± 18.4 and 233.7 ± 10.4 s after diazepam 0.1 and 1.0 mg/kg, respectively ($F = 5.081$; df = 2,14; $P < 0.022$). Concomitant with this was a significant reduction in inactivity. After administration of vehicle, separation resulted in 67.0 ±

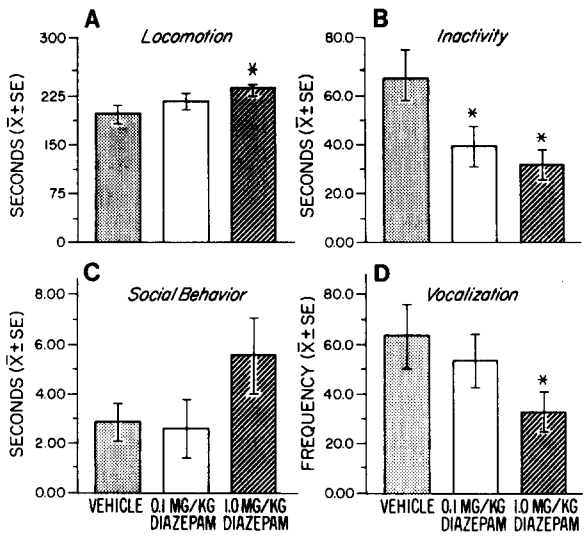


Fig. 1. Effects of diazepam on the amount of separation-induced behavior. A: locomotion (s per test session). B: inactivity (s per test session). C: social behavior (s per test session). D: vocalizations (frequency per test session). *Differs from results for vehicle, $P < 0.05$.

10.9 s of inactivity, as compared to 39.3 ± 9.4 and 32.2 ± 6.5 s after 0.1 and 1.0 mg/kg diazepam ($F = 8.999$; df = 2,14; $P < 0.003$). Even though the conditions of separation markedly limited the ability of the animals to socialize, a significant increase in social behavior was seen after diazepam. Social behavior was rated when one animal was within 15 cm of another animal or when the monkeys manually contacted each other through the mesh separating the cages. Administration of vehicle and diazepam 0.1 mg/kg resulted in similar amounts of social behavior, 2.9 ± 0.6 and 2.6 ± 1.0 s. Diazepam 1.0 mg/kg increased the amount of social behavior to 5.6 ± 1.2 s ($F = 3.871$; df = 2,14; $P < 0.046$).

Distress vocalizations have been used in many studies to characterize the intensity of the separation response in primates[19,20,40]. Diazepam significantly reduced the frequency of these separation-induced vocalizations. After vehicle there were 63.0 ± 16.6 vocalizations, compared to 53.3 ± 14.5 and 32.6 ± 10.4 after diazepam 0.1 mg/kg and 1.0 mg/kg ($F = 5.342$; df = 2,14; $P < 0.019$). Clearly, the effect of diazepam on vocalizations was not a result of sedation, as activity levels increased after administration of diazepam.

Table II shows the results of Expt. 2. None of the behaviors rated were significantly affected by either dose of Ro 15-1788, although changes in locomotion, inactivity, and self-mouth were close to reaching statistical significance. Ro 15-1788 5 mg/kg decreased locomotion and increased inactivity and self-mouth, whereas at the 10 mg/kg dose a small increase in locomotion was seen along with a decrease in inactivity.

In general, the results of Expt. 3 confirmed the findings of Expts. 1 and 2. Vehicle/Ro 15-1788 did not markedly affect any of the behaviors. Diazepam/vehicle non-significantly increased locomotion to 257.0 ± 8.4 s, compared to 228.9 ± 15.2 s after vehicle/vehicle. Repeated measures ANOVAs revealed significant effects of the drug treatments on inactivity ($F = 3.126$; df = 3,21; $P < 0.048$) and on vocalizations ($F = 6.963$; df = 3,21; $P < 0.002$) (Fig. 2). As in Expt. 1, diazepam decreased inactivity and distress vocalizations (Fig. 2A). In this experiment, social behavior was unaffected by diazepam. Ro 15-1788 reversed the effects of diazepam on inactivity, but did not alter diazepam's effects on the frequency of vocalizations (Fig. 2B).

TABLE II

Effects of Ro 15-1788 on the behavioral response of infants to brief maternal separation

Data are means ± S.E.M.s. Behaviors are scored in seconds per 5-min test session, except for vocalizations, which are expressed as frequency per 5-min test session.

Behavior	*Drug*			*P*
	Vehicle	*Ro 15-1788 5 mg/kg*	*Ro 15-1788 10 mg/kg*	
Locomotion	212.0 ± 15.7	184.1 ± 13.4	228.9 ± 5.2	< 0.058
Inactivity	55.1 ± 9.0	72.8 ± 9.5	39.3 ± 7.4	< 0.066
Environmental exploration	24.1 ± 7.2	22.0 ± 7.8	22.3 ± 6.2	n.s.
Vocalizations	56.4 ± 16.4	47.1 ± 13.7	51.1 ± 14.8	n.s.
Self-mouth	3.0 ± 1.8	14.1 ± 6.0	6.6 ± 4.7	< 0.082
Social	3.1 ± 1.0	3.2 ± 0.9	2.2 ± 1.9	n.s.

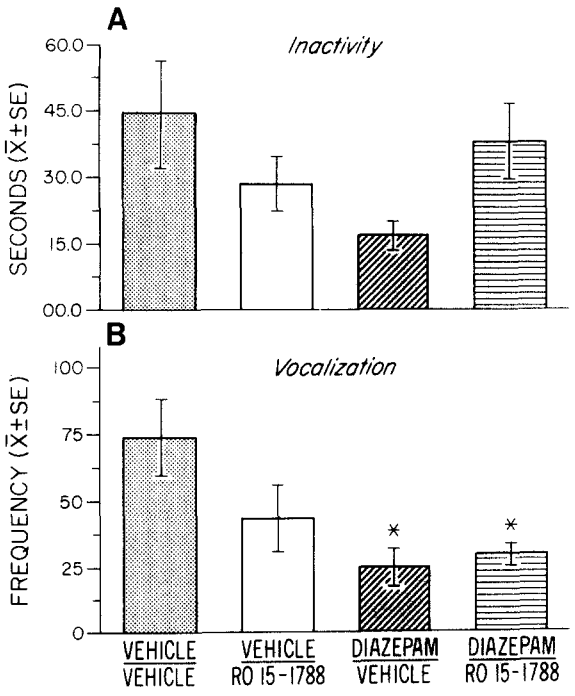


Fig. 2. Effects of diazepam 1.0 mg/kg followed by Ro 15-1788 10 mg/kg on separation-induced behavior. A: inactivity (s per test session). B: vocalizations (frequency per test session). *Differs from results for vehicle/vehicle, $P < 0.05$.

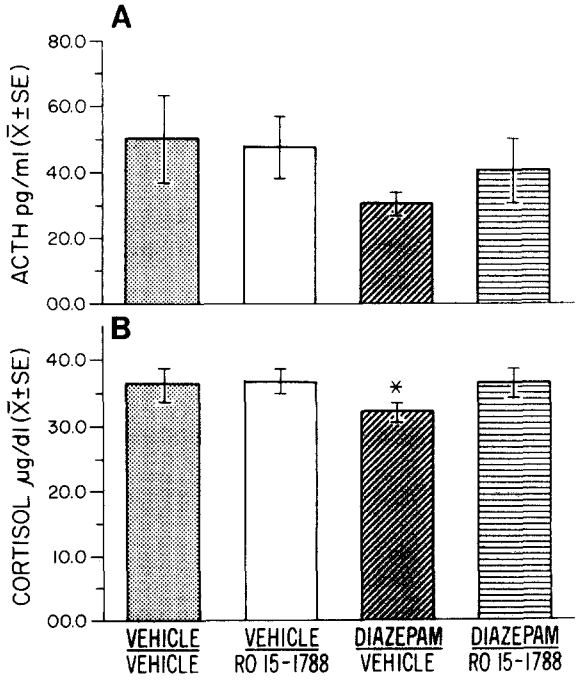


Fig. 3. Effects of diazepam 1.0 mg/kg followed by Ro 15-1788 10 mg/kg on plasma concentrations of (A) ACTH and (B) cortisol. *Differs from results for vehicle/vehicle, $P < 0.05$.

Fig. 3 (A, B) demonstrates the effects of the 4 treatments on the separation-induced secretion of ACTH and cortisol. Although changes in the ACTH concentrations with the different treatments did not reach statistical significance, the changes were in the predicted direction. After vehicle/vehicle, the mean ± S.E.M. ACTH concentration was 50.0 ± 13.37 pg/ml and decreased to 30.6 ± 3.4 pg/ml after diazepam/vehicle. After diazepam/Ro 15-1788, the ACTH concentration returned to 40.3 ± 9.69. A significant treatment effect was found for the cortisol concentrations ($F = 3.066$; df = 3,21; $P < 0.05$). While the changes were small in magnitude, the concentration of cortisol after the diazepam/vehicle treatment was less than that for all other treatments.

DISCUSSION

Our results demonstrate that the benzodiazepine agonist diazepam alters the behavioral and endocrine responses of infant rhesus monkeys separated from their mothers. In the first experiment diazepam increased activity and social behavior and decreased distress vocalizations. In the second experiment, Ro 15-1788 did not significantly affect the monkey's behavior. However, it is of interest that the low dose of Ro 15-1788 had effects that could be considered anxiogenic, consistent with File's finding that low doses of Ro 15-1788 have anxiogenic effects in rats[13]. In other experiments with peer-separated monkeys, we found that the inverse agonist β-carboline-3-carboxylic acid ethyl ester had a profile of effects similar to that of low-dose Ro 15-1788 (unpublished data). The effects of Ro 15-1788 10 mg/kg were in the same direction as the effects of diazepam, and other investigators[16] have found that high doses of this compound have anxiolytic effects in rodent models. Whether the intrinsic effects of the low and high doses of this compound are mediated by a direct effect on the benzodiazepine receptor, or whether its intrinsic actions occur by blocking the effects of endogenously released ligands acting at the benzodiazepine receptor, has not yet been determined.

In Expt. 3, we demonstrated that Ro 15-1788 blocked the diazepam-induced decrease in inactivity. This is clear evidence that this effect of diazepam is mediated through benzodiazepine receptors. Lack of a similar effect on diazepam-induced decrease in vo-

calizations suggests the possibility that the decrease in vocalizations is mediated through non-benzodiazepine systems. However, in work with older monkeys separated from their peers, we demonstrated that Ro 15-1788 5 mg/kg blocked the increase in vocalizations induced by β-carboline-3-carboxylic acid ethyl ester (unpublished data). To demonstrate unequivocally whether diazepam's effects on vocalizations are mediated by benzodiazepine receptors or by another mechanism, further work should be done varying the dose and time of administration of Ro 15-1788 in relationship to diazepam.

The endocrine data follow the behavioral data in that the highest levels of ACTH and cortisol were seen after vehicle/vehicle and the lowest after diazepam/vehicle, and diazepam/Ro 15-1788 reversed the effects of diazepam/vehicle on these hormones. Since ACTH is secreted episodically and has a short plasma half-life, it is possible that more frequent blood sampling over the course of the experiment would have revealed greater effects of diazepam on ACTH concentrations. That diazepam's effect on activity and the pituitary–adrenal hormones was reversed by Ro 15-1788 suggests that these effects are mediated through benzodiazepine receptors. Overall, our data support the hypothesis that in primates, endogenous benzodiazepine systems modulate the behavioral and endocrine response to naturally occurring stressors.

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